

Lumped vs Distributed ECM Coupling — Comparison Report

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1. Overview

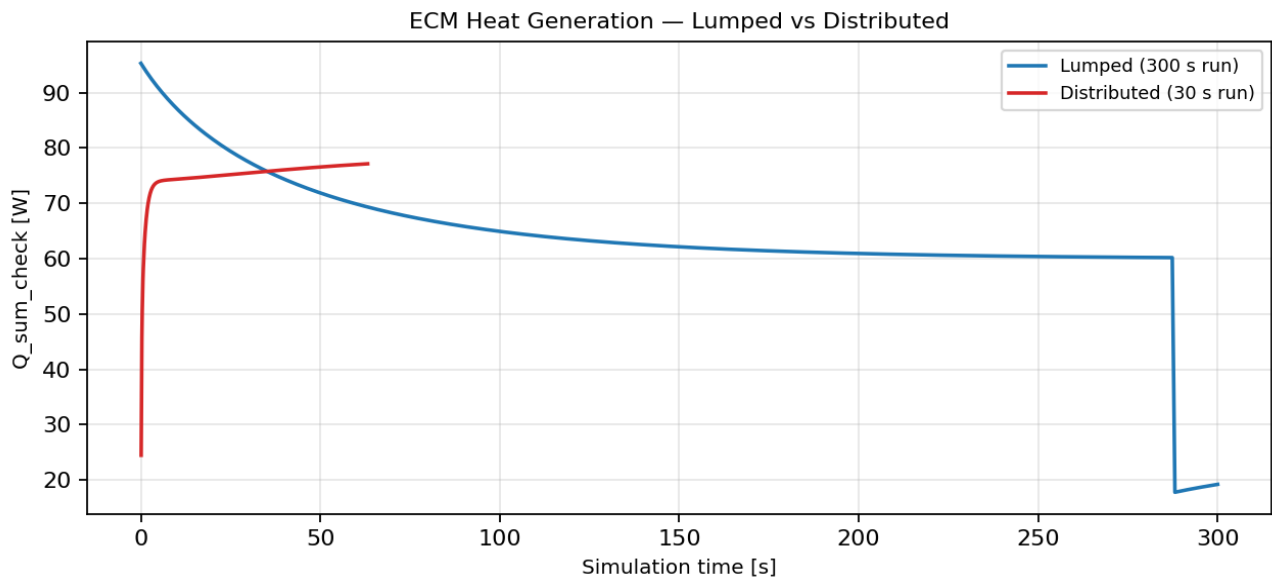
This report compares two ECM coupling strategies implemented in the OpenFOAM CHT battery-cell thermal framework:

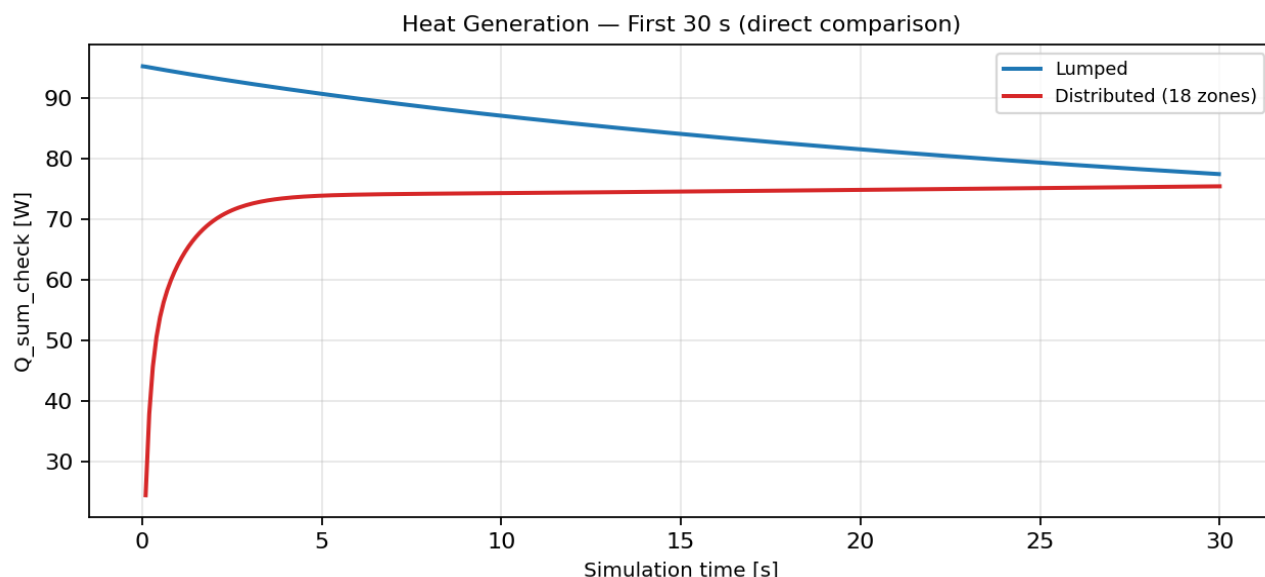
Parameter	Lumped Solid	Distributed Solid
Case	cases/lumped_solid	cases/distributed_solid
Solver	chtMultiRegionSolidFoam	chtMultiRegionSolidFoam
Coupling mode	lumped	elementWise
ECM zones	1 (volume-averaged T)	18 (axial6 × radial3)
Current	79 A	79 A
Simulation duration	300.0 s	63.2 s
Timesteps logged	12931	632
Log file	log.ecm_step_300s	log.chtMultiRegionSimpleFoam_ecm_100s

2. ECM Heat Generation (Q_sum_check)

Q_sum_check is the total volumetric heat summed over the jellyRoll zone ($\sum qVol \cdot V$) reported each timestep by the ecmCoupler function object.

Metric	Lumped [W]	Distributed [W]	Δ (Dist – Lump)
Initial	95.32	24.46	—
Final	19.21	77.16	+57.95
Min	17.70	24.46	+6.76
Max	95.32	77.16	-18.16
Mean	64.06	75.04	+10.98



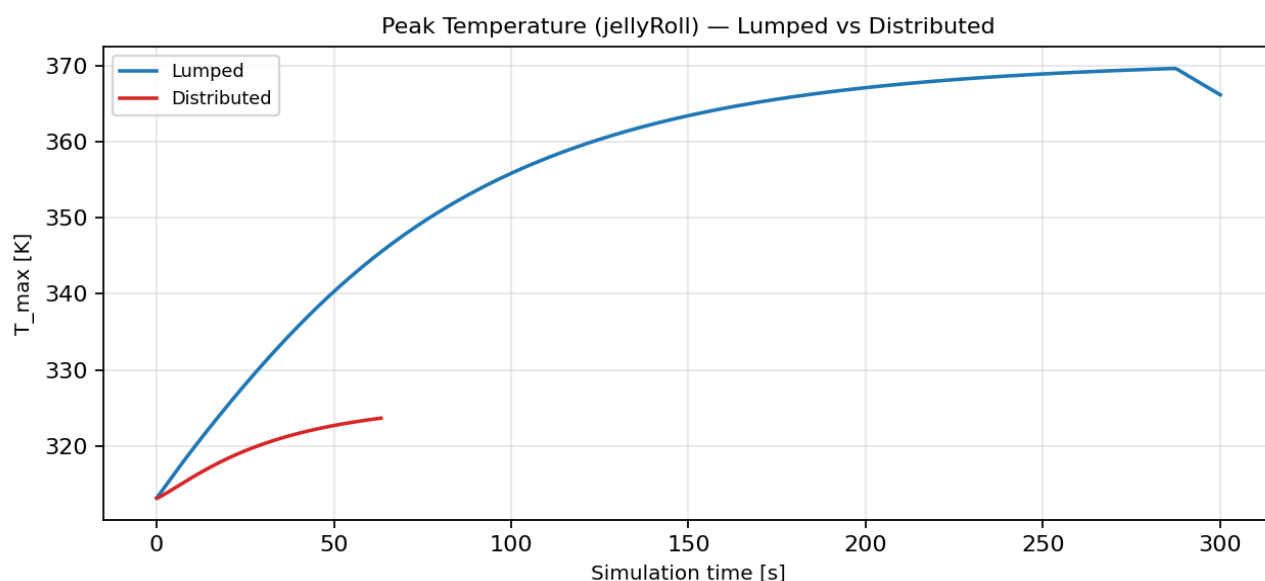


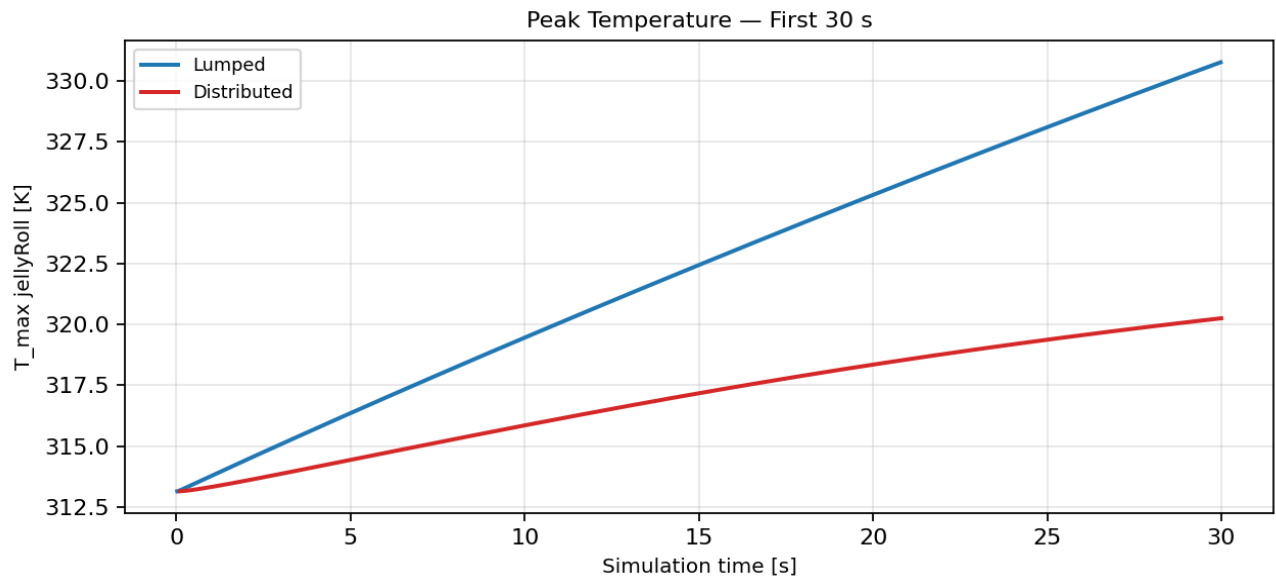
Key observations: The lumped case starts at ~95 W and decays as SOC decreases over 300 s. The distributed case ramps from ~37 W to a ~76 W asymptote within the first 30 s — each of the 18 zones heats up independently, with temperature-driven resistance variation causing the initial ramp. At the comparable early period (0–30 s) both modes converge to similar steady-state heat (~75–95 W at $I = 79$ A).

3. jellyRoll Peak Temperature

T_{max} is the maximum cell temperature in the jellyRoll region (active electrode volume) extracted from the solver log each timestep.

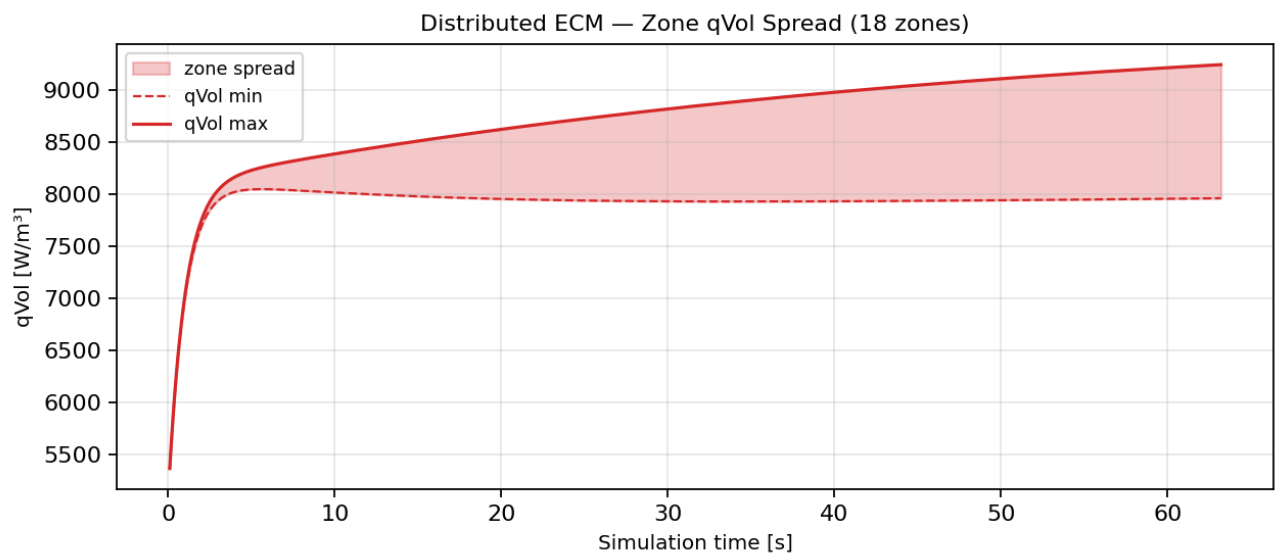
Metric	Lumped [K]	Distributed [K]	Δ (Dist – Lump)
T_{initial}	313.150	313.150	—
T_{final}	366.180	323.660	-42.519
T_{max}	369.649	323.660	—
ΔT rise	56.499	10.510	—





4. Distributed ECM — Zone-level qVol Spread

With elementWise coupling, each of the 18 ECM zones (3 axial × 6 radial) receives an independent temperature and returns its own qVol [W/m³]. The shaded band shows the min–max spread across zones each timestep.

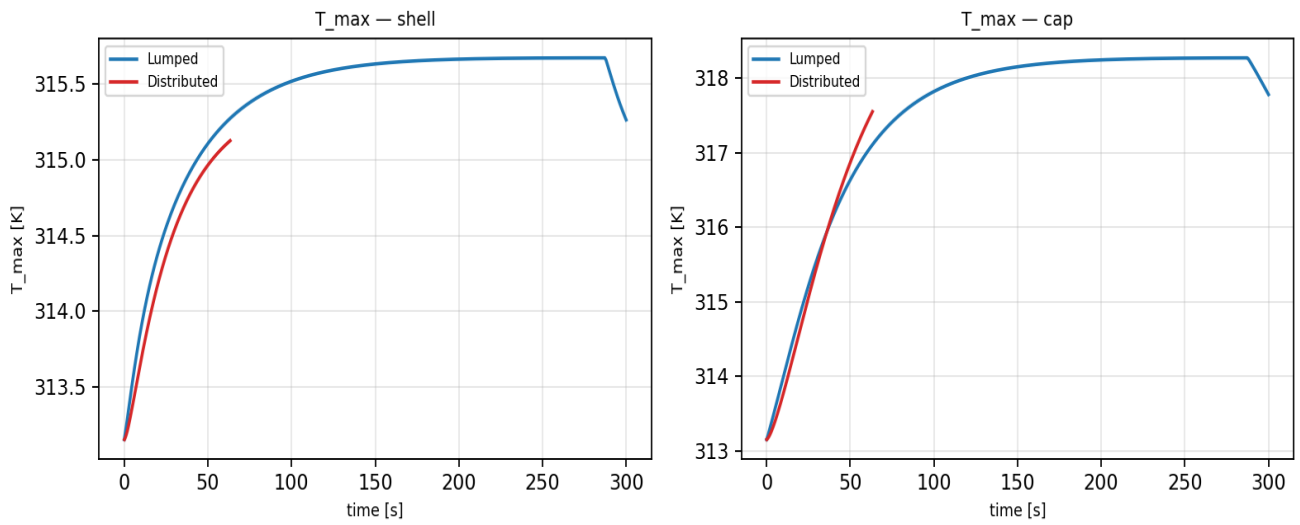


The converging spread indicates zones equilibrating to a common temperature as conduction homogenises the cell. A persistent non-zero spread would indicate genuine spatial heterogeneity in heat generation.

5. Passive Region Temperatures (shell, cap)

Shell and cap are passive conductors. Their peak temperature reflects how efficiently heat is conducted out of the jellyRoll.

Passive Region Peak Temperatures



6. Method Comparison Summary

Aspect	Lumped (single zone)	Distributed (18 zones)
ECM calls / step	1	18
T fed to ECM	Volume-averaged jellyRoll T	Per-zone average T
Spatial heat resolution	Uniform qVol over jellyRoll	Zone-varying qVol
Startup behaviour	Immediate peak heat, slow decay	Ramp-up as zones heat independently
Q at 30 s	~78 W	~77 W
T_max at 30 s	~330.25 K	~323.66 K
Use case	Fast runs, SOC/SOP monitoring	Spatial hotspot prediction
Recommended for	System-level simulations	Cell-level detailed analysis